

Cancer Investigation

ISSN: 0735-7907 (Print) 1532-4192 (Online) Journal homepage: www.tandfonline.com/journals/icnv20

Low Cancer Screening Rates Among Emergency Department Patients: A Multicenter Retrospective Study

Nicholas R. Pettit, Matthew Hays, Xiaochun Li, Jacquiline Wiltshire & Paul Musey

To cite this article: Nicholas R. Pettit, Matthew Hays, Xiaochun Li, Jacquiline Wiltshire & Paul Musey (2026) Low Cancer Screening Rates Among Emergency Department Patients: A Multicenter Retrospective Study, Cancer Investigation, 44:5, 555-558, DOI: 10.1080/07357907.2026.2618576

To link to this article: https://doi.org/10.1080/07357907.2026.2618576

View supplementary material

Published online: 02 Feb 2026.

Submit your article to this journal

Article views: 53

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=icnv20

CANCER INVESTIGATION
2026, VOL. 44, NO. 5, 555–558
https://doi.org/10.1080/07357907.2026.2618576

Taylor & Francis
Taylor & Francis Group

BRIEF REPORT

Low Cancer Screening Rates Among Emergency Department Patients: A Multicenter Retrospective Study

Nicholas R. Pettit[a], Matthew Hays[b], Xiaochun Li[b], Jacquiline Wiltshire[b] and Paul Musey[a]

[a]Department of Emergency Medicine, Indiana University School of Medicine, Indianapolis, IN, USA; [b]Indiana University School of Medicine and Richard M Fairbanks School of Public Health, Indianapolis, IN, USA

ABSTRACT

Many Americans, especially emergency department (ED) patients, are not up to date with routine cancer screenings. This multi-center retrospective study assessed adherence to U.S. Preventive services task force guidelines for breast, colorectal, and lung cancer screenings among 198,210 patients across 11 Indiana EDs. Results showed 33.0% were screened for colorectal cancer, 55.4% of eligible females for breast cancer, and only 3.1% of eligible patients for lung cancer. These findings highlight persistently low screening rates in ED populations. Targeted interventions are needed to improve adherence and reduce disparities in cancer prevention among high-risk groups.

ARTICLE HISTORY
Received 1 May 2025
Revised 17 December 2025
Accepted 12 January 2026

KEYWORDS
Cancer; Cancer screening; Preventive health

Introduction

People who lack reliable access to healthcare, including cancer screening, are more likely to be diagnosed at advanced stages of cancer, when treatment is less effective (1). Emergency departments (EDs) tend to serve in a safety-net capacity for many patients (1). EDs may be a target-rich environment for interventions aimed at increasing the awareness and uptake of cancer screening (2). Current studies assessing the frequency at which ED patients are appropriately screened for colorectal cancer (CRC), breast and lung cancers are limited. The objective of this study, is to assess the rate at which ED patients are appropriately screened for CRC, breast, and lung cancers based on cancer screening guidelines set forth by the United States Preventive Services Task Force (USPSTF) within the largest healthcare system in Indiana (3).

Methods

This is a multi-center (11 EDs), retrospective, cohort study of ED patients aged 50–80 from 1/1/2019 until 1/1/2023. Sites included two level 1 trauma centers, 3 critical access EDs, and six suburban community EDs, collectively representing the largest healthcare system in Indiana. This work is in compliance with the Strengthening the Reporting of Observational studies in Epidemiology (STROBE) guidelines. De-identified patient data was abstracted (via single data pull) from the electronic health record (EHR) including age, gender, race, ethnicity, insurance payer, smoking status (yes/no), and cancer screening adherence based on USPSTF criteria (3). USPSTF eligibility criteria and diagnostic tests used to assess for cancer screening within the EHR are included in Supplementary Table 1. Continuous variables were expressed as median with interquartile ranges. Categorical variables are presented as *n* (%) and were compared with the use of Pearson's chi-square test. Multivariable logistic regression was performed with smoking status, race, sex, and Medicaid status as covariates to model odds of receiving various screenings based on USPSTF guidelines.

CONTACT Nicholas Pettit NRPettit@iu.edu Department of Emergency Medicine, Indiana University School of Medicine, 720 Eskenazi Ave, 3rd Floor FOB Building, Indianapolis IN 46202, USA.

Supplemental data for this article can be accessed online at https://doi.org/10.1080/07357907.2026.2618576.

© 2026 Taylor & Francis Group, LLC

Results

This study included 198,210 adult ED patients (Table 1). Patients were divided based on smoking status, given smoking is a required risk factor for LCS eligibility. Overall, the study population included 103,656 (52.3%) females, however higher frequencies of men were observed in the smoking cohort (53.0% smokers vs 42.7% nonsmokers). Table 2 presents the cancer screening outcomes for eligible patients in this cohort. Focusing on CRC screening, 59,376 adults (59,376/179,882, 33.0%) had received CRC screening. Meanwhile, 51,726 females (55.5% of the eligible cohort) had received breast cancer screening. Lastly, only 2,982 (3.0%) of eligible adults had received a low dose computed tomography (LDCT) for LCS. However, 9,691 adults (10.0%) did receive some sort of outpatient chest computed tomography (non-LDCT).

Multivariable analysis predicting the odds of receiving age and risk-factor based cancer screenings is included in Table 3. For both breast and colorectal cancer screenings, smoking status was associated with increased odds of having received cancer screening (breast cancer screening OR 95% CI, colorectal cancer screening OR 95% CI. For both CRC screening and LCS, black race and other race were associated with lower rates of cancer screening attendance, comparatively where black patients had increased odds of having attended BCS. Males were less likely to receive guideline concordant screenings for both breast

Table 1. Study population.

	Total	Smokers ($N = 96608$)	Nonsmokers ($N = 101602$)
Age, mean (SD), median (Q1–Q3)	62.41 (8.61), 61 (55–69)	62.39 (8.56), 61 (55–69)	62.43 (8.66), 61 (55–69)
Sex at birth			
Female	103,656	45,438	58,218
Male	94,554	51,170	43,384
Race			
White	157,606	77,904	79,702
Black or African American	31,431	16,435	14,996
Other	4,257	1,001	3,256
Missing/unknown/refused	4,445	1,139	3,306
Multiracial	471	129	342
Ethnicity			
Hispanic/Latino	10,396	2,607	7,789
Not Hispanic/Latino	185,150	93,143	92,007
Unknown	2,664	858	1,806
Insurance type			
Auto/TPL	7,150	3,633	3,517
COVID19	512	219	293
Commercial	18,736	9,381	9,355
Guaranteed acct	4,749	2,930	1,819
HIP	11,573	6,783	4,790
Incarcerated	1,319	1058	261
Managed care	71,253	30,470	40,783
Medicaid	19,645	12,475	7,710
Medicare	100,509	53,854	46,655
Other	1,001	474	527
Other government	26,904	16,478	10,426
Self pay	41,118	20,414	20,704
Special contracts	1,344	841	503
Workers' comp	4,943	2,278	2,665

Table 2. Frequency at which emergency departments are adherent to cancer screening.

	Total (total eligible, %)	Smokers (total eligible, %)	Nonsmokers (total eligible, %)	Odds ratio[b] (95% CI)
Received colorectal cancer screening	59,376 (179,882, 33.0%)	32,573 (87,782, 37.1%)	26,803 (92,100, 29.1%)	1.42 (1.39–1.45)
Received breast cancer (eligible females)	51,726 (93,275, 55.4%)	24,232 (41,269, 58.7%)	27,494 (52,006, 52.9%)	1.27 (1.24–1.31)
Received outpatient lung cancer screening (LDCT[a])	3,126[c]	2,982 (96,614, 3.1%)	144 (101,624, 0.1%)	33.69 (28.91–39.26)
Received outpatient computed tomography scan (non-LDCT)	13,808[c]	9,691 (96,614, 10.0%)	4,117 (101,624, 4.1%)	2.45 (2.40–2.50)

[a]LDCT: Low dose computed tomography.
[b]Smokers to nonsmokers.
[c]Total eligible are not included in this column since smoking is a requirement for cancer screening.

Table 3. Multivariable regression model for receiving appropriate cancer screening.

Variable	Odds ratio	95% confidence interval
Breast cancer screening ($n = 91,351$)		
Smoker	1.257	1.225 to 1.291
Black Race[a]	1.43	1.104 to 1.183
Multi Race[a]	0.986	0.758 to 1.286
Other Race[a]	1.197	1.095 to 1.308
Medicaid Insurance[b]	0.867	0.840 to 0.913
Colorectal cancer screening ($n = 175,664$)		
Smoker	1.427	1.398 to 1.457
Black Race[a]	0.860	0.837 to 0.885
Multi Race[a]	0.604	0.479 to 0.761
Other Race[a]	0.875	0.814 to 0.940
Male[c]	0.754	0.739 to 0.770
Medicaid Insurance[b]	0.873	0.843 to 0.903
Lung cancer screening ($n = 140,538$)		
Black Race[a]	0.497	0.440 to 0.561
Other Race[a]	0.422	0.273 to 0.652
Male[c]	0.746	0.693 to 0.803
Medicaid[b]	1.244	1.114 to 1.388

[a]Reference group: white race.
[b]Reference group: non-Medicaid insurance.
[c]Reference group: female.

and colorectal cancers (OR 95% CI, OR 95% CI, breast and colorectal respectively).

Discussion

In this study, we found that most ED patients did not receive guideline recommend cancer screening for breast, CRC, and lung cancers. Sixty seven percent of adults did not have CRC screening, 44.5% of eligible females did not have breast cancer screening, and 97.0% of smokers did not receive LCS. Comparatively, data from The American Lung Association has reported that in 2023 and 2024 the rate of LCS among those living in Indiana was 5.1% and 16.9%, respectively, both of which are well above the 3% LCS rate observed within this study (4). Our study represents the largest cohort study evaluating the rate of cancer screening adherence among ED patients. One similar study enrolled 221 patients, also demonstrating poor rates of cancer screening attendance for CRC, breast, and cervical cancers (1). Another study identified the ED was among the most frequent points of contact for patients eligible for LCS, with 49% ($n = 225$) of the 450 eligible patients in the study being seen in the ED (5).

Most notable is the poor rate of LCS attendance among ED smokers. While 10% of the LCS-eligible cohort did receive outpatient chest CT scans, they were the incorrect diagnostic test for LCS and likely either represent those patients are not in our LCS-program (which order LDCT) or their clinical indications were not for cancer screening. Lung cancer is the most deadly cancer, and despite the demonstrable reduction in mortality with LCS, rates of LCS attendance remain in the single digits (6). While some studies have assessed barriers to screening, future research should focus on interventions for at-risk patients. One study has already demonstrated feasibility and efficacy of referring ED patients for LCS upon ED discharge demonstrating the ED could be a key setting for educating and referring patients for cancer screenings (1,7). Other opportunities include looking at functional genomics and deep learning methods for cancer prediction and identification of at-risk patients who would benefit from LCS (8).

Multivariate analysis demonstrated that men were less likely to receive LCS and CRC screening compared to women. Additionally, individuals of black race had lower rates of LCS and CRC screening compared to individuals of white race. These findings are consistent with prior literature showing that men consistently underutilize preventive health care services, relative to women, including cancer screening (9). Meanwhile, black individuals experience a disproportionate cancer burden, characterized by higher cancer mortality and persistently lower screening rates compared to white individuals (10) Understanding racial and sex differences in cancer screening and preventive health care utilization is crucial for informing future ED-based interventions. Black patients often report higher ED utilization and are more likely to lack a usual source of care, which may present opportunities to adapt care delivery by selectively targeting preventive health interventions to high-risk populations during the ED encounter (11).

In conclusion, our findings demonstrate that within Indiana's largest healthcare network, most ED patients appear to be poorly adherent to nationally accepted cancer screening recommendations. Future work should explore ED-based interventions aimed to identify patients with screening needs, engage with patients regarding cancer screening, and provide an effective

method to conduct or connect patients with cancer screening.

Acknowledgements

Regenstrief Institutes, Inc.'s participation in this project is acknowledged for access to and expertise in data management.

Authors contributions

NP and PM conceived the study. NP, PM, JW obtained funding. MH and XL provided statistical advice and analyzed the data. NP drafted the manuscript, and all authors contributed substantially to its revisions. NP takes responsibility for the paper as a whole.

Disclosure statement

The authors declare no conflicts of interest.

Funding

American Cancer Society CDSG-23-1025372-01-CPSH, IU Simon Cancer Center Program Pilot Junior Faculty Award

Data sharing statement

Partial or complete datasets and data dictionary are available upon request to Dr. Nicholas Pettit (nrpettit@iu.edu) to investigators who provide an IRB letter of approval.

References

1. Heinert SW, Ahmed M, Guzman-Baez K, Penugonda A, Oh S, Aamir A, et al. Characteristics and barriers of emergency department patients overdue for cancer screening. West J Emerg Med. 2024;25(1):101–110. doi:10.5811/westjem.60400.
2. Adler D, Abar B, Chiao EY. Emergency department-based cancer screening interventions. Emerg Cancer Care. 2022;1(1):3. doi:10.1186/s44201-022-00003-8.
3. Force USPST. A&B Recommendations. 2024. [cited 2024 Sep 19]. Available from: https://www.uspreventiveservicestaskforce.org/uspstf/recommendation-topics/uspstf-a-and-b-recommendations.
4. Association AL. State of Lung Cancer Indiana. 2024. [2025 May 28]. Available from: https://www.lung.org/research/state-of-lung-cancer/states/indiana.
5. Kaltman M, Yalamanchili S, Park JA, Weyant MJ, Suzuki K. High catchment areas for lung cancer screening-eligible patients. Ann Thorac Surg. 2026;121(1):181–185. doi:10.1016/j.athoracsur.2025.04.029.
6. Adams SJ, Stone E, Baldwin DR, Vliegenthart R, Lee P, Fintelmann FJ. Lung cancer screening. Lancet. 2023;401(10374):390–408. doi:10.1016/S0140-6736(22)01694-4.
7. Pettit NR, Horner D, Freeman S, Rieger K. Pilot intervention to increase uptake of lung cancer screening through the emergency department. Am J Emerg Med. 2024;79:157–160. doi:10.1016/j.ajem.2024.02.019.
8. Tran TO, Vo TH, Le NQK. Omics-based deep learning approaches for lung cancer decision-making and therapeutics development. Brief Funct Genomics. 2024;23(3):181–192. doi:10.1093/bfgp/elad031.
9. Davis JL, Buchanan KL, Katz RV, Green BL. Gender differences in cancer screening beliefs, behaviors, and willingness to participate: implications for health promotion. Am J Mens Health. 2012;6(3):211–217. doi:10.1177/1557988311425853.
10. Saka AH, Giaquinto AN, McCullough LE, Tossas KY, Star J, Jemal A, et al. Cancer statistics for African American and Black people, 2025. CA Cancer J Clin. 2025;75(2):111–140. doi:10.3322/caac.21874.
11. Parast L, Mathews M, Martino S, Lehrman WG, Stark D, Elliott MN. Racial/ethnic differences in emergency department utilization and experience. J Gen Intern Med. 2022;37(1):49–56. doi:10.1007/s11606-021-06738-0.